

Model Compression using Pruning, Quantization, and Distillation on AG News Dataset

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Assignment: Assignment 5 – Model Compression

Objective

Compare pruning, quantization, and knowledge distillation on a small transformer model trained on AG News. Measure the impact of each technique on model size, validation accuracy, and inference latency (time per batch) on CPU.

Motivation

Transformer models offer strong performance but are costly in storage and compute. By evaluating compression techniques, this study aims to identify methods to reduce resource usage while preserving accuracy, enabling deployment on memory- and compute-constrained devices.

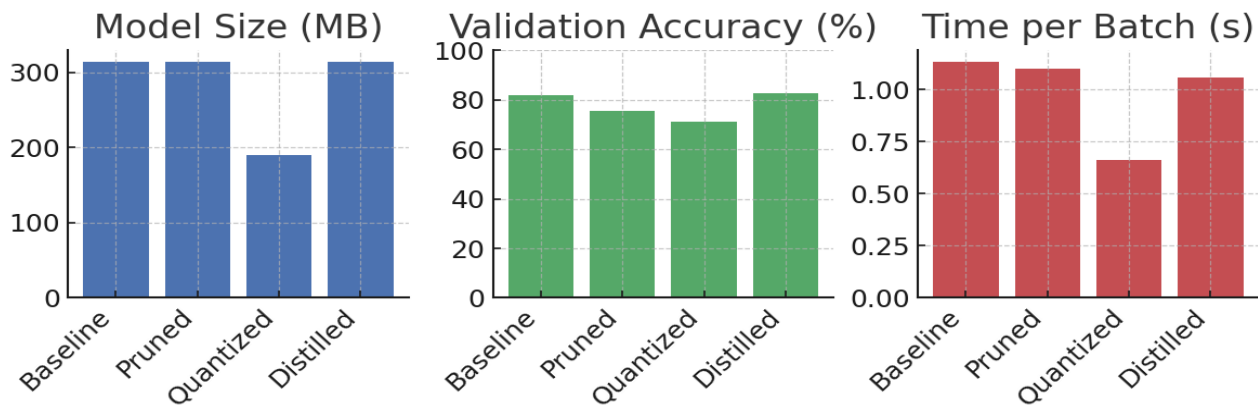
Experimental Methodology

Dataset: AG News (subset of 1,000 samples).
Baseline model: small transformer fine-tuned on AG News.
Compressors: Global L1 pruning (30%), Dynamic INT8 quantization, Knowledge Distillation (teacher → student).
Evaluation: Model size (MB), Validation accuracy (%), Time per batch (s) on CPU.

Results

Model Variant	Size (MB)	Val Accuracy (%)	Time/Batch (s)
Baseline	313.30	82.0	1.1326
Pruned	313.30	75.5	1.1033
Quantized	190.13	71.0	0.6613
Distilled	313.30	82.5	1.0591

Comparison Plots



Key Observations

- Quantization provides the largest reduction in model size and the fastest per-batch inference time on CPU, but with a modest drop in validation accuracy compared to the baseline.
- Pruning reduced computation but in this run did not reduce stored model size due to framework serialization specifics; accuracy declined moderately.
- Knowledge distillation produced a student with comparable accuracy to the baseline while allowing potential future model-size reductions through student architecture design.
- For deployment to edge devices, quantization offers the best immediate gains in size and speed, while distillation is promising when training a smaller student that better fits device constraints.

Conclusion

This comparative study shows that different compression techniques trade off size, speed, and accuracy in distinct ways. Dynamic quantization is an effective first step to reduce memory footprint and accelerate CPU inference. Pruning can help in reducing computations and may require framework-aware serialization to gain storage benefits. Knowledge distillation remains the most flexible approach to obtain compact models tailored for deployment when combined with appropriate student architectures.

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